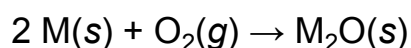


Subject:
General Chemistry

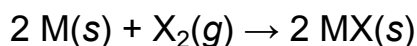
System:
Atomic Theory and Chemical Composition

The alkali metals are a highly reactive family of metals occupying Group 1 (1A) of the periodic table. In their elemental form, the alkali metals have a shiny, metallic appearance, but they are also very soft and can be easily cut by hand. Due to their high reactivity, however, the alkali metals do not exist in their elemental form in nature.

Because the alkali metals possess only a single valence electron with a low first ionization energy, the alkali metals (M) readily react with nonmetals and lose this electron to form cations with a +1 charge (M^+) that are incorporated into ionic compounds. Consequently, elemental samples of alkali metals will rapidly tarnish in oxygen by forming oxides (Reaction 1) and vigorously react with elemental halogens (X_2) to form halide salts (Reaction 2).

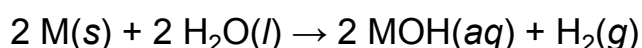


Reaction 1



Reaction 2

Alkali metals also react violently with water to form very basic metal hydroxide (alkali) solutions while releasing hydrogen gas in the process (Reaction 3).



Reaction 3

As demonstrated by these reactions, alkali metals possess a low electron affinity that makes accepting an additional electron to form an anion less favorable than losing the valence electron to form a cation.

Of the alkali metals, francium is the rarest since it is a trace radioactive byproduct formed from the decay of uranium ores and has a very short half-life. As a result, any francium formed will quickly decay into radium or astatine by beta decay or by alpha decay, respectively.

Whereas hydrogen is a part of Group 1 based on its number of valence electrons, hydrogen is not an alkali metal. Interestingly, some research has suggested that under extreme pressures and very low temperatures, hydrogen may form a solid with metallic properties.

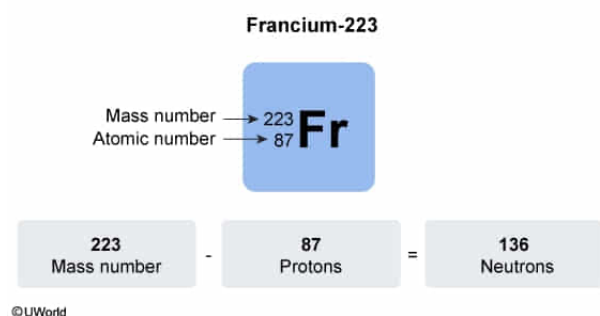
What is the number of neutrons in an atom of francium-223?

- ☐ A. 87 [7%]
- ✓ ☒ B. 136 [88%]
- ☐ C. 223 [3%]
- ☐ D. 310

Correct

89% answered correctly

Explanation:



Isotopes are atoms of the *same element* that have a different **number of neutrons** in the nucleus. Therefore, **isotopes** are designated by their **mass number** (the sum of the number of **protons** and the **neutrons** in the nucleus) in one of two ways: by using the element name hyphenated with its mass number (e.g., francium-223), or by using a chemical symbol with the mass number given as a superscript to the upper left (e.g., ^{223}Fr).

Because the **atomic number** (the **number of protons**) determines the *identity* of an element and is the same for all atoms of that element, the number of protons is not always explicitly stated in isotope notation; however, when stated it is placed as a subscript to the lower left of the chemical symbol (below the mass number). By knowing the name (or symbol) of an element, the number of protons can be determined by referencing the periodic table. An atom of any isotope in its elemental state will be **neutral** (have **no net charge**), and the number of negatively charged electrons will be equal to the number of positively charged protons.

For an isotope of francium-223, the mass number (protons plus neutrons) is 223. Because all atoms of francium have 87 protons, the number of neutrons is found by subtracting the atomic number (protons) from the mass number ($223 - 87 = 136$).

(Choice A) The atomic number (the number of protons) is 87.

(Choice C) The isotope mass number (the sum of the number of protons and neutrons) is 223.

(Choice D) The total number of particles in the isotope (protons, neutrons, and

electrons) is 310.

Educational objective:

Isotopes are designated by their mass number, which is the sum of the number of protons and neutrons in the nucleus. The number of neutrons in an isotope can be determined by subtracting the element's atomic number (number of protons) from the mass number. The number of electrons in a neutral atom is equal to the number of protons.

Time spent:0 sec

Subject:

General Chemistry

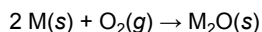
QID:401649

System:

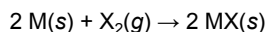
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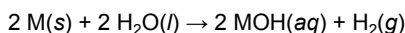


Reaction 1



Reaction 2

Alkali metals also react violently with water to form very basic metal hydroxide (alkali) solutions while releasing hydrogen gas in the process (Reaction 3).



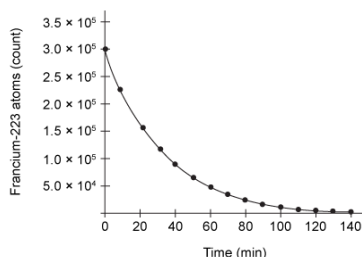
Reaction 3

As demonstrated by these reactions, alkali metals possess a low electron affinity that makes accepting an additional electron to form an anion less favorable than losing the valence electron to form a cation.

Of the alkali metals, francium is the rarest since it is a trace radioactive byproduct formed from the decay of uranium ores and has a very short half-life. As a result, any francium formed will quickly decay into radium or astatine by beta decay or by alpha decay, respectively.

Whereas hydrogen is a part of Group 1 based on its number of valence electrons, hydrogen is not an alkali metal. Interestingly, some research has suggested that under extreme pressures and very low temperatures, hydrogen may form a solid with metallic properties.

The graph below illustrates the radioactive decay of a population of francium-223 atoms. How long did it take for two half-lives of francium-223 to elapse?

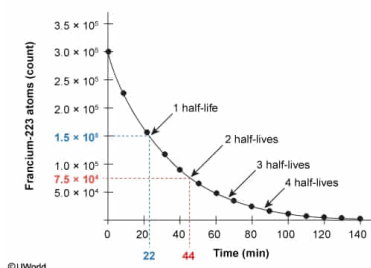


- ☐ A. 20 min [3%]
- ☒ B. 44 min [89%]
- ☐ C. 65 min [4%]
- ☐ D. 90 min [1%]

Correct

89% answered correctly

Explanation:



In a **radioactive decay**, the **half-life** is the time required for the amount of a given isotope to **decrease by half**. With each subsequent half-life interval that passes, the amount remaining again decreases by half and the fraction of the initial amount that remains becomes

exponentially smaller, eventually approaching zero. Accordingly, the fraction remaining after n half-lives can be calculated:

Fraction

After one half-life passes ($n = 1$), the fraction remaining is half of the initial amount. Following a second half-life interval ($n = 2$), the fraction remaining is again cut in half, making the resulting amount equal to one-fourth the initial amount. Three half-lives ($n = 3$) would leave an amount equal to one-eighth the initial amount.

Taken from the graph at time zero, the initial amount of francium-223 (^{223}Fr) was 3.0×10^5 atoms. Following two half-life intervals ($n = 2$), the fraction remaining of the ^{223}Fr atoms should be one-fourth the initial amount (leaving 7.5×10^4 atoms). From the graph, the time required to reach this amount is seen to be 44 minutes.

This question can also be answered from an alternative perspective: When the initial amount dropped by one-half to 1.5×10^5 atoms, 22 minutes had elapsed. This means that the half-life of ^{223}Fr is 22 minutes. Therefore, two half-lives equal 44 minutes.

(Choice A) A span of 20 minutes is a little less than one half-life, leaving nearly half the initial amount of ^{223}Fr .

(Choice C) A span of 65 minutes is a little less than three half-lives, leaving nearly one-eighth the initial amount of ^{223}Fr .

(Choice D) A span of 90 minutes is a little over four half-lives, leaving a little less than one-sixteenth the initial amount of ^{223}Fr .

Educational objective:

The half-life is the time required for the amount of a given radioactive isotope to decrease by half. With each subsequent half-life interval that passes, the amount that remains decreases by half again, becoming exponentially smaller and eventually approaching zero.

Time spent: 0 sec

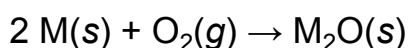
Subject:
General Chemistry

QID: 401650

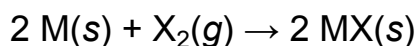
System:
Atomic Theory and Chemical Composition

The alkali metals are a highly reactive family of metals occupying Group 1 (1A) of the periodic table. In their elemental form, the alkali metals have a shiny, metallic appearance, but they are also very soft and can be easily cut by hand. Due to their high reactivity, however, the alkali metals do not exist in their elemental form in nature.

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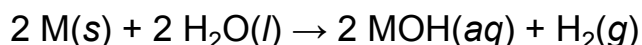


Reaction 1



Reaction 2

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Reaction 3

As demonstrated by these reactions, alkali metals possess a low electron affinity that makes accepting an additional electron to form an anion less favorable than losing the valence electron to form a cation.

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Whereas hydrogen is a part of Group 1 based on its number of valence electrons, hydrogen is not an alkali metal. Interestingly, some research has suggested that under extreme pressures and very low temperatures, hydrogen may form a solid with metallic properties.

If elemental hydrogen forms a metallic solid under extreme pressure, as mentioned in

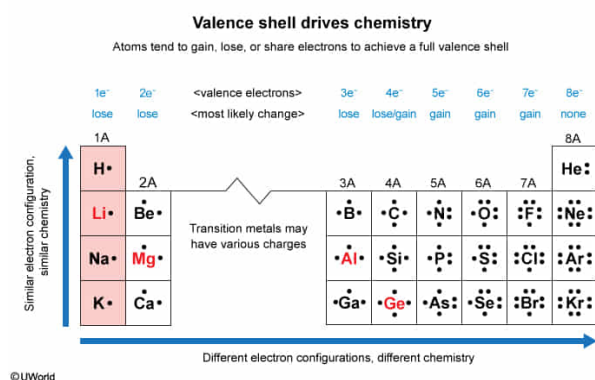
the passage, the chemical properties of the metallic hydrogen formed should be similar to those of which element?

- ☐ A. Germanium [4%]
- ☐ B. Aluminum [7%]
- ☐ C. Magnesium [5%]
- ✓ ☒ D. Lithium [82%]

Correct

82% answered correctly

Explanation:



An atom is more stable when it has an **electron configuration** that results in a **full valence shell** (outer shell). As a result, atoms are driven to participate in chemical reactions and form bonds either by **electron transfers** (gaining or losing electrons) or by **electron sharing** in order to achieve a more stable configuration with a full valence shell.

In the **representative elements** (except hydrogen and helium), a **full valence shell** is usually achieved by eight valence electrons (an octet). For atoms with 1-3 valence electrons, it is energetically more favorable to **lose electrons** and retain the fully occupied lower shells than to gain additional electrons to complete an octet in the partially filled higher shell. Conversely, for atoms with 5-7 electrons, gaining the **additional electrons** for an octet is more favorable than losing several electrons.

Because **valence electrons** drive the chemistry of an atom, **atoms in the same group** (column) of the periodic table express **similar chemical properties** (bonding, reactivity, etc.) whereas atoms in different groups display different chemical properties. Hydrogen and lithium are both in Group 1A and have **1 valence electron** each. On this basis, if hydrogen were to form a metallic solid under extreme pressures, it would be expected to have chemical properties more similar to those of lithium than to those of other groups' elements.

(Choices A, B, and C) Magnesium (Group 2A, 2 valence electrons), aluminum (Group 3A, 3 valence electrons), and germanium (Group 4A, 4 valence electrons) all

have more valence electrons than the elements of Group 1A (1 valence electron). As a result, the chemical properties of magnesium, aluminum, and germanium should be very different from those of metallic hydrogen.

Educational objective:

The number of valence electrons drives chemical behavior. Atoms tend to gain, lose, or share electrons to achieve a full valence shell. Elements within the same group of the periodic table have the same number of valence electrons and exhibit similar chemical properties.

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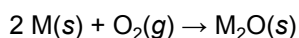
Subject:
General Chemistry

QID:401651

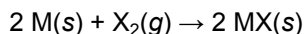
System:
Atomic Theory and Chemical Composition

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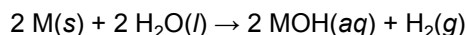


Reaction 1



Reaction 2

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Reaction 3

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Whereas hydrogen is a part of Group 1 based on its number of valence electrons, hydrogen is not an alkali metal. Interestingly, some research has suggested that under extreme pressures and very low temperatures, hydrogen may form a solid with metallic properties.

According to trends in the first ionization energy, which of the following alkali metals is the LEAST reactive?

- ✓ ☒ A. Li [54%]
☐ B. Na
☐ C. K
☐ D. Cs [44%]

Correct

55% answered correctly

Explanation:

Selected chemical properties of alkali metals

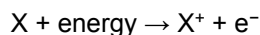
Group 1		Group 2
H		
Li	Be	
Na	Mg	
K	Ca	
Rb	Sr	
Cs	Ba	

Ion formation (oxidation)
 $M \rightarrow M^+ + e^-$

Reaction with water
 $2 M(s) + 2 H_2O(l) \rightarrow 2 MOH(aq) + H_2(g)$

Increasing reactivity

The **first ionization energy** is the energy required to **remove the first**, most loosely bound **valence electron** (e^-) from a neutral atom (X). It can be represented generally as:



Elements with loosely bound valence electrons have lower first ionization energies (i.e., are easier to ionize) than elements with tightly bound valence electrons. On the periodic table, the first ionization energy tends to **increase across a period** (row), but it tends to **decrease moving down a group** (column) with some intermittent exceptions.

Alkali metals occupy Group 1 and have the lowest **first ionization energies** of all the element groups. As the atoms in the alkali metal group **increase in size** moving down the column, the **valence electron** occupies an increasingly **higher level** that is farther from the nucleus (ie, experiences less attraction) and is less tightly bound. As a result, less ionization energy is required to remove the electron from the valence shell of larger alkali metals, which makes them more reactive.

Lithium is the **smallest atom** in Group 1 and has a valence electron in a **lower level** with a greater attraction to the nucleus (ie, the electron takes more energy to remove). Therefore, **lithium** is the **least reactive** of the alkali metals in Reactions 1-3.

(Choices B, C, and D) Sodium (Na), potassium (K), and cesium (Cs) are all alkali metals, but each has a lower ionization energy than lithium (Li). For reactions with nonmetals, less energy is required to remove an electron from Na, K, or Cs than from Li, making the larger alkali metals more reactive than Li.

Educational objective:

The first ionization energy is the energy required to remove the first, most loosely bound valence electron from a neutral atom. The first ionization energy tends to increase across a period and decrease moving down a group on the periodic table. Elements with a lower first ionization energy are easier to ionize and more reactive than elements with a higher ionization energy.

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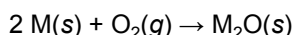
Subject:
General Chemistry

QID: 401652

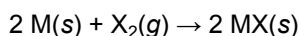
System:
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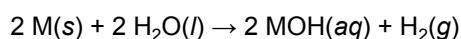


Reaction 1



Reaction 2

Alkali metals also react violently with water to form very basic metal hydroxide (alkali) solutions while releasing hydrogen gas in the process (Reaction 3).



Reaction 3

As demonstrated by these reactions, alkali metals possess a low electron affinity that makes accepting an additional electron to form an anion less favorable than losing the valence electron to form a cation.

Of the alkali metals, francium is the rarest since it is a trace radioactive byproduct formed from the decay of uranium ores and has a very short half-life. As a result, any francium formed will quickly decay into radium or astatine by beta decay or by alpha decay, respectively.

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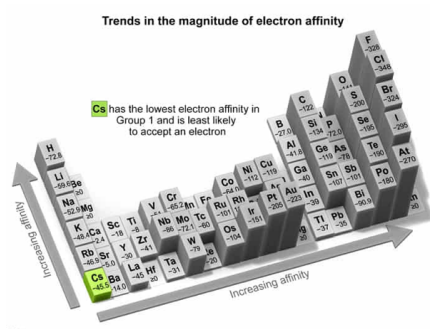
Which of the following alkali metals is LEAST likely to accept an additional electron?

- ☐ A. Na [28%]
- ☐ B. K
- ☐ C. Rb
- ✓ ☒ D. Cs [70%]

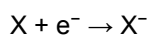
Correct

70% answered correctly

Explanation:



Electron affinity is defined as the **change in energy** resulting from **adding an electron** (e^-) to a neutral atom of an element (X) in the gas state to form an anion with a -1 charge:



When a stable anion is formed, energy is released as a product and a negative value results for the change in energy (energy of the products minus energy of the reactants) from the process. Therefore, electron affinity values with a **more negative magnitude** indicate elements that **more readily accept** the addition of an **electron**. In general, an additional electron is often more easily **accepted by nonmetals** than by metals. Periodic trends show that apart from some intermittent, regional exceptions caused by atomic size and orbital-filling effects, electron affinity tends to become **stronger** (more negative) moving left to right **across a period** and tends to become **weaker** (less negative) moving **down a group** on the periodic table.

Consistent with these general trends, the electron affinity of the alkali metals in Group 1 weakens moving down the column. Therefore, cesium (Cs) has the weakest electron affinity and is the least likely (least favorable) to accept an additional electron.

(Choices A, B, and C) Sodium (Na), potassium (K), and rubidium (Rb) are all alkali metals, but each has a stronger electron affinity than cesium (Cs). Adding an additional electron to Na, K, or Rb forms a more stable anion and releases more energy than is released from Cs.

Educational objective:

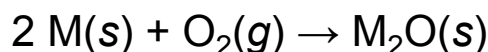
Electron affinity is the change in energy associated with adding an electron to a neutral atom in the gas state. Values with a more negative magnitude indicate elements that more readily accept the addition of an electron. With some intermittent exceptions, electron affinity tends to become stronger (more negative) moving left to right across a period and tends to become weaker (less negative) moving down a group on the periodic table.

Time spent: 0 sec
Subject:
General Chemistry

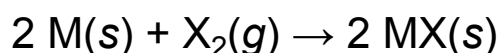
QID: 401653
System:
Atomic Theory and Chemical Composition

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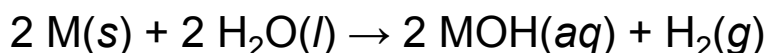


Reaction 1



Reaction 2

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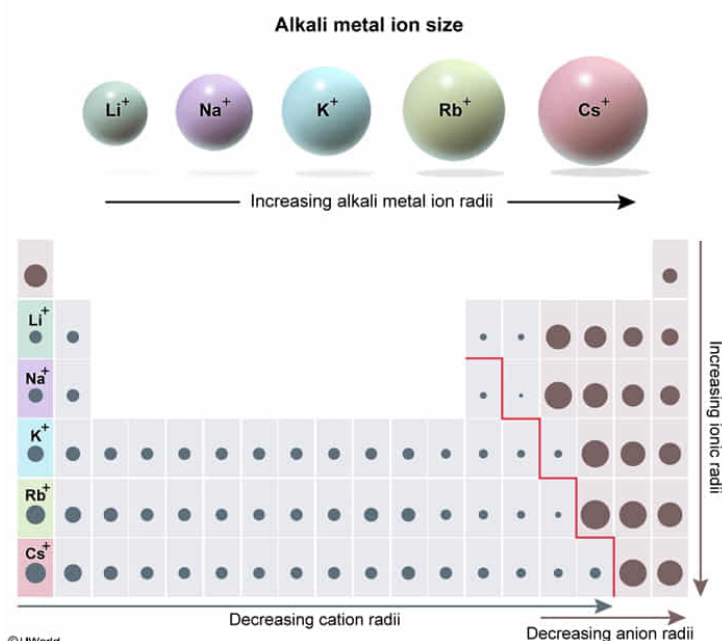
At higher concentrations, alkali metal cations with an ionic radius close to that of a potassium ion can displace potassium ions in biological systems and promote low potassium levels (hypokalemia). Which cation is most likely to displace K^+ ions?

- ☐ A. Li^+ [10%]
- ✓ ☒ B. Rb^+ [86%]
- ☐ C. Cs^+ [1%]
- ☐ D. Fr^+ [1%]

Correct

87% answered correctly

Explanation:



Ionic radii tend to **decrease across a period** (row) of the periodic table (left to right) and **increase** moving **down a group** (column). This trend occurs for metal cations, and then resets and repeats for anions beginning near the division between **metals and nonmetals**, past which anions tend to preferentially form. Losing electrons to form a cation causes the remaining electrons to experience a **greater effective nuclear charge (Z_{eff})**, pulling the electrons closer to the nucleus.

Conversely, gaining electrons to form an anion produces greater electronic repulsion and nuclear shielding (lesser Z_{eff}), which pushes electrons farther from the nucleus.

Alkali metal ions that are similar in size (ie, have a comparable ionic radius) to K^+ ions are the most likely to displace a K^+ ion. Because the ionic radii of the alkali metal ions increase moving down the column, the **elements adjacent** to potassium will form ions that are **closer** to the **size** of the K^+ ion. As a result, out of the options given, a Rb^+ ion is closer to the size of a K^+ ion and will be more likely to displace K^+ ions than Li^+ , Cs^+ , or Fr^+ ions.

(Choice A) Because alkali metal ions decrease in size moving up the column, Li^+ is much smaller than K^+ .

(Choices C and D) Because Cs^+ and Fr^+ are larger than Rb^+ (and even larger than K^+), these ions do not displace K^+ as effectively as Rb^+ .

Educational objective:

For ions having the same charge, ionic radii tend to decrease across a period and increase down a group on the periodic table. Ions of adjacent elements tend to be closer in size than those farther away.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401654
System:
Atomic Theory and Chemical Composition